

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 10	Team No -	Project Title - Online Code Judge
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1. Abstract

This project presents a scalable, distributed Online Code Judge platform designed to support the full lifecycle of competitive programming from problem authoring to real-time contest evaluation. Problem setters can compose challenges with structured, multi-case test suites, while contestants submit solutions across multiple programming languages including C++, Python, Java, and JavaScript through a browser-based Monaco Editor interface.

At the core of the system is a sandboxed judge engine powered by Docker, which spins up an isolated container per submission and enforces strict CPU, memory, and execution time constraints, ensuring that malicious or runaway code cannot affect the host environment. Submissions enter a Redis-backed BullMQ job queue and are asynchronously dispatched to judge workers, which evaluate code against predefined test cases and return structured verdicts - Accepted, Wrong Answer, Time Limit Exceeded, Memory Limit Exceeded, Runtime Error, or Compilation Error - persisted in MongoDB with full submission metadata.

Contest sessions are managed through a dedicated Contest Service that tracks participant activity, enforces time boundaries, and maintains real-time leaderboards backed by Redis sorted sets, enabling $O(\log n)$ rank updates at scale. A plagiarism detection module employs AST-based diffing and token fingerprinting to identify suspiciously similar submissions within a contest window.

The backend follows a microservices architecture built on Node.js and Express.js, with an API Gateway, Judge Worker, and Contest Management service communicating asynchronously over the Redis queue. MongoDB serves as the primary persistent store, with schema design optimized for leaderboard aggregation pipelines and submission history queries across large datasets.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090169	Bharagava Raaj Vamsi	
2520090020	Tarun	
2520090107	Hanish Kumar	

3. Frameworks / Platforms / Software Used

The frontend is built using React.js integrated with the Monaco Editor (the engine behind VS Code) to provide a rich, browser-based code editing experience. The backend is developed in Node.js with Express.js, structured as microservices covering the API Gateway, Judge Worker, and Contest Management. MongoDB serves as the primary database handling persistent storage for users, problems, submissions, and contest data. Redis is used as both a job queue (via BullMQ) for managing submission dispatch and as an in-memory store for real-time leaderboard ranking using sorted sets. The judge engine leverages Docker to spin up isolated containers per submission, enforcing strict CPU, memory, and time constraints during code execution. Plagiarism detection is implemented through AST-based diffing and token fingerprinting techniques. The entire system is containerized and orchestrated using Docker Compose for consistent local and production deployments.

Guide Name and Signature (remarks if any) _____